

Question number	Answer	Additional guidance	Mark
7(a)(i)	A description that includes the following points: • substitution mutations {swap / change / replaces} a {base / nucleotide} (1) • insertion and deletion mutations (results in a) change in the number of {bases / nucleotide} (1)	ACCEPT plural ACCEPT increases and decreases number of bases / greater or fewer number of bases / more or less bases changes length of {DNA / gene} / length of DNA gets longer or shorter IGNORE frameshift	(2)

Question number	Answer	Additional guidance	Mark
7(a)(ii)	An explanation that includes four of the following points: • substitution mutations will not affect protein function (1) • as substitution mutations may not alter the {amino acid / amino acid sequence} (1) • as the substitution mutation may occur in the non-coding region (of the DNA) (1) • substitution mutations may not affect protein function (1) • as substitution mutations may change one amino acid (only) (1)	ACCEPT {works / activity} for function ACCEPT may code for the same {amino acid / amino acid sequence} ACCEPT {works / activity} for function	(4)

	• {deletion / insertion} mutations may have (large) effects on protein function (1) • as {insertion / deletion} mutations cause a frameshift (1)	ACCEPT {works / activity} for function ACCEPT description of frameshift alters the sequence of amino acids after the mutation	
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Question number	Answer	Additional guidance	Mark
7(a)(iii)	• mouse: more animals sampled (1)	ACCEPT easier to control variable NB answers about sample size must be comparative	(1)

Question number	Answer	Additional guidance	Mark
7(b)(i)	An answer that includes the following points: • an appropriate line of best fit (1) • whole number between 285 and 1143 (1)	straight line (reasonable if drawn by hand) does not extend further than the left hand or right hand points touches data for all 4 age groups	(2)

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Question number	Answer	Additional guidance	Mark
7(b)(ii)	A description that includes the following points: • number (of mutations) increases with age (in all four animals) (1) • different species have different life spans (1) • different species have different number of mutations (in their cells) (1) • different species have different {mutation rates / number of mutations per cell per year} (1) • humans have the greatest variation and naked mole rats have the least variation (in the number of mutations per cell) (1)	ACCEPT different species have different frequency of mutations	(3)

Question number	Answer	Additional guidance	Mark
7(b)(iii)	An answer that includes the following points: • cells divide at different {rates / frequencies} (1) • different {repair mechanisms / DNA polymerases} (1) • exposed to different {factors (that cause mutations) / environment} 1) • different {quantities of DNA / base sequences / genes} (1)	ACCEPT named example of a cell type ACCEPT named example e.g. UV ACCEPT alleles / chromosomes genetic predisposition / genetics	(2)